DIGITAL EVIDENCE CHAIN-OF-CUSTODY TRACKER

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INTRODUCTION

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The Digital Evidence Chain-of-Custody Tracker is a blockchain-based system designed to securely record and monitor the transfer of evidence during investigations. By using immutable smart contracts and wallet-based authentication, the system ensures that every evidence movement is permanently logged and verifiable. This prevents tampering, increases transparency, and strengthens the credibility of evidence presented in court.

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Literature Review

Title of Paper	Aim of the Paper	Limitations Identified
[1] Blockchain-Based Chain of Custody for Digital Forensics – M. Conti, A. Dehghantanha, K. R. Choo	To provide a secure and tamper-proof mechanism for tracking digital evidence using blockchain in forensic investigations	Primarily conceptual; lacks real-time implementation and integration with existing systems
[2] A Blockchain-Based Framework for Secure Evidence Management – Z. Zhang, X. Chen	To develop a decentralized system for storing and verifying digital evidence securely	High storage cost when storing large data on-chain; performance issues in large-scale systems
[3]Blockchain for Digital Forensics: Opportunities and Challenges – M. Conti, S. Kumar Karuppayah	To analyze how blockchain can improve reliability and transparency in digital forensics	Focuses on theoretical analysis; lacks practical deployment and performance evaluation

PROBLEM DEFINITION

& OBJECTIVES

Evidence tampering and manipulation of chain-of-custody records can compromise investigations and weaken legal proceedings.

Objectives-

- **Create unique Evidence IDs**
- **Log every transfer immutably**
- **Authenticate officers securely**
- **Provide court-verifiable audit trail**

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Proposed Solution

1. **The system uses a hybrid architecture combining Firebase for efficient storage and blockchain for secure evidence verification, with AI handling capture and categorization.**
2. **Each file is hashed using SHA-256 and stored on blockchain, ensuring any tampering can be detected.**
3. **Blockchain enables a secure chain-of-custody, with QR codes linking physical evidence to digital records.**

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Scope

1. Smart contract stores Evidence ID & transfer history
2. Each officer uses blockchain wallet authentication
3. Only evidence hashes stored on-chain
4. Backend (Node.js) interacts with blockchain
5. QR code links physical evidence to digital record
6. Deploy on Polygon testnet / private Ethereum

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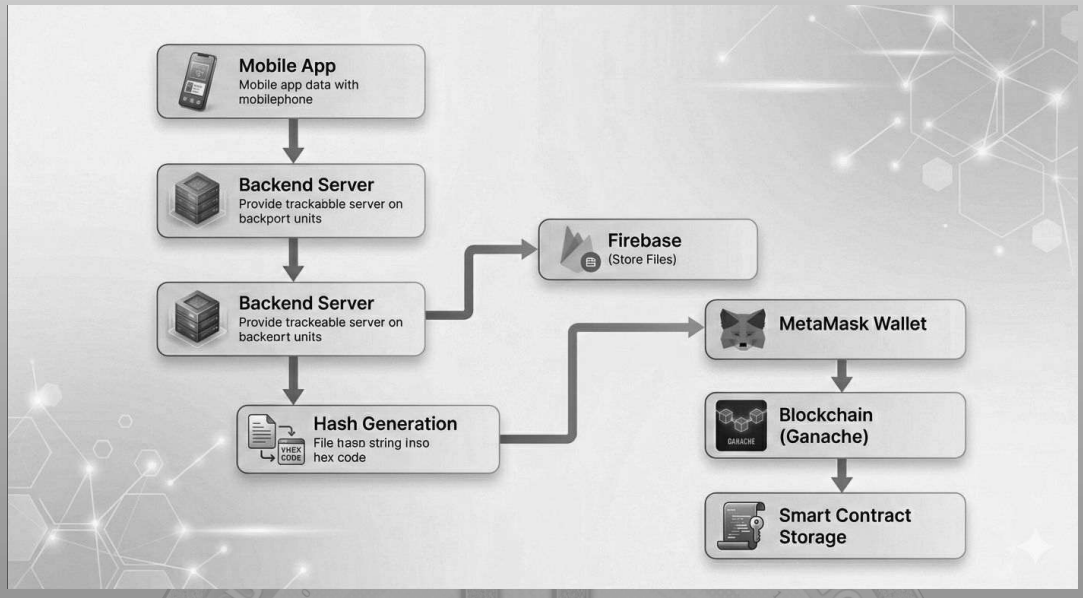
System Design



Hardware & Software

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- **Remix IDE – Smart contract development & testing**
- **Solidity – Smart contract programming**
- **MetaMask – Officer wallet authentication**
- **Ganache – Local private Ethereum blockchain**
- **Node.js + Express – Backend integration**
- **MongoDB – Metadata storage (off-chain)**
- **QR Code Generator – Physical evidence tagging**

Implementation

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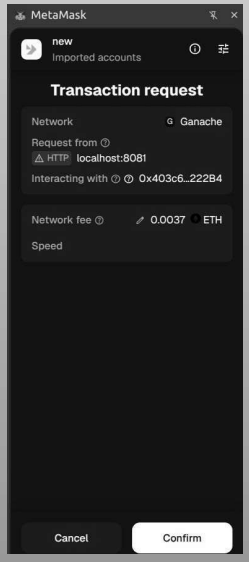
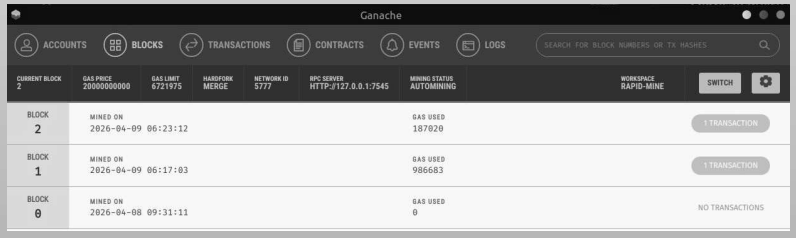


Fig 1



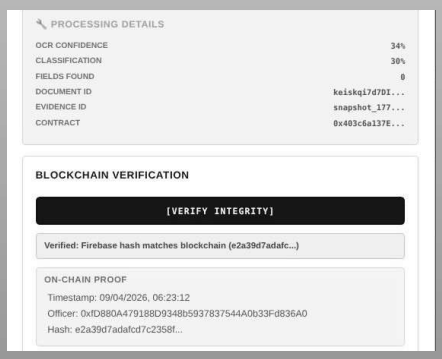


Fig 2

Fig 3

The project's blockchain flow hashes each document, writes the hash and evidence ID to the EvidenceStorage.sol contract via BlockchainService.ts, and verifies later by comparing the on-chain hash with Firebase and/or a fresh image hash

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CONCLUSION

In conclusion, the Digital Evidence Chain-of-Custody Tracker leverages blockchain technology to create a secure, transparent, and tamper-proof system for managing investigative evidence, ensuring trust, accountability, and legal reliability. This system demonstrates a practical real-world application of blockchain in strengthening institutional integrity and digital security.

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References

[1] M. Conti, A. Dehghantanha, and K. R. Choo, "Blockchain-based chain of custody for digital forensics," IEEE Transactions on Information Forensics and Security, vol. 15, pp. 1–12, 2020.

[2] Z. Zhang and X. Chen, "A blockchain-based framework for secure evidence management," IEEE Access, vol. 7, pp. 12345–12356, 2019.

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[4] H. Tian, J. He, and Y. Ding, "A secure data sharing system based on Hyperledger Fabric blockchain," Future Generation Computer Systems, vol. 95, pp. 88–98, 2019.